

PRODUCT CATALOG · POWER & WORKSHOP

EN760 Series Intelligent Frequency Inverters

Variable-speed drives for fans and water pumps, 0.75 kW to 500 kW, 380 V three-phase. Constant-pressure water supply, multi-pump control and built-in PID — direct from the factory to your farm.



WHO WE ARE

Farm equipment, at the price it *actually* costs.

Built by farmers, for farmers. Andrivo sources farm infrastructure and equipment from international manufacturers, handles the entire import process, and delivers to your farm anywhere in Southern Africa.

Farmers usually pay for a chain of markups they never see — a factory, an export agent, an importer and a local distributor, each adding a margin nobody discloses. Andrivo removes that chain. Our only charge is a service fee on top of the true cost: 30%, or 25% with a referral code.

25%

fee with a referral code (30% without)

0

distributors in between

SADC

we deliver across Southern Africa

How every quote works

1**Factory direct**

We work directly with the manufacturer of every machine in this catalog — no middlemen.

2**Inspected before it ships**

Each order is checked against the factory spec sheet before it leaves the plant.

3**Shipping & customs handled**

Freight, duties, clearing and transport to your gate are all managed by us.

4**Parts & aftersales**

Spares in stock, and every machine logged to your farm under My Farm by Andrivo.

CONTENTS

What's in this catalog

Specifications are typical for each model; your Andrivo quote confirms the exact model, size and price for your pump, fan or motor.

Design features

Precise industry positioning	04
Compact structural design	04
Built-in application macros	04
Operation keypad	04

Performance & applications

Multi-mode water supply	05
Dedicated fan & pump functions	05
High reliability	05
Motor types, PID & energy saving	06

Protection features	06
---------------------	----

I/O & communication	06
---------------------	----

Technical specifications

Input, output & control performance	07
-------------------------------------	----

I/O channels, environment & structure	08
---------------------------------------	----

Models & sizes

Model naming guide	09
--------------------	----

Model, current & motor table	09
------------------------------	----

Dimensions & mounting	10
-----------------------	----

Busbar assemblies & keypads	11
-----------------------------	----

Design Features



Precise industry positioning

Precise industry positioning: Specifically designed for quadratic torque loads (i.e., load torque is proportional to the square of the speed) such as pumps, fans, and compressors. This makes it perform exceptionally well in fields such as HVAC (Heating, Ventilation and Air Conditioning), water supply, and wastewater treatment.



Compact structural design

Its compact design saves control cabinet space. It supports side-by-side installation, which helps to improve the utilization of cabinet space.



Built-in application macros

A variety of application macros are pre-programmed. Users only need to select the corresponding macro and make a few parameter settings to quickly put it into operation, which greatly simplifies the debugging process.



Operation Keypad

Equipped with an easy-to-use control panel with potentiometers (knobs), it allows for quick parameter setting, start/stop, and speed adjustment.



- Optional features include a rotary keypad with external communication access (EN-LED15-D, EN-LED16-D).



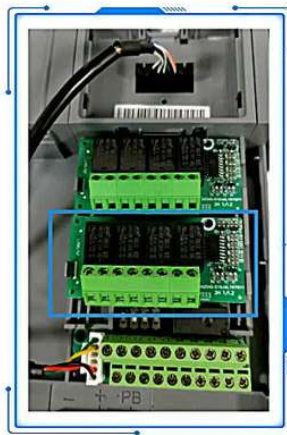
Performance characteristics



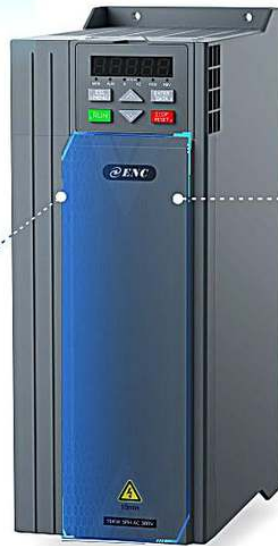
Multi-mode water supply

- Single-unit water supply: one Frequency Inverter drives one motor to provide constant pressure water supply.

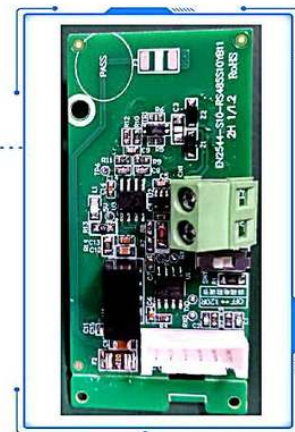
- It supports one-to-many water supply, supporting up to one-to-four water supply, realizing the logic of one Frequency Inverter driving four motors to switch between industrial frequency and variable frequency for water supply.



- ✓ EN-WS9 Constant Pressure Water Supply Four-Channel Relay Expansion Card: One card for dual-channel constant pressure water supply; two cards for quad-channel constant pressure water supply.



- The Networked water supply system uses a star topology network structure to centrally control up to 5 Frequency Inverters driving 5 motors, thereby realizing the constant pressure water supply logic function.



- ✓ EN-TX485-76 Expansion Card (Adds one more 485 terminal interface; this card is compatible with the entire series)

Application characteristics



Dedicated functions for fans and water pumps

- **Multi-pump control function:** Some models support the control of multiple pumps, enabling alternating operation, sequential start-stop, and pump switching, balancing the operating time of each pump, and extending the system life.
- **Automatic acceleration and deceleration:** Automatically adjusts acceleration and deceleration times according to load characteristics to avoid water hammer effect (for pump systems) and pressure fluctuations.



High reliability

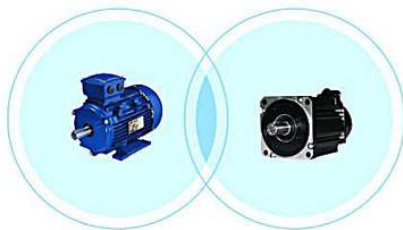
- Coated circuit boards can resist moisture, corrosive gases and dust, improving reliability in harsh environments.
- The wide voltage range design provides good adaptability to power grid fluctuations.

Performance characteristics



Supports asynchronous and synchronous motors

It supports speed control for three-phase asynchronous motors and three-phase permanent magnet synchronous motors.



Optimized PID controller

It features a built-in advanced PID controller, making it particularly suitable for applications requiring constant pressure, flow rate, or temperature (such as constant pressure water supply and ventilation systems). It automatically adjusts the output frequency based on actual feedback values to achieve precise control.



Outstanding energy-saving characteristics

Sleep/Wake-up Function : When demand is low (such as at night), the inverter can automatically stop running and automatically start when demand recovers, further saving energy.



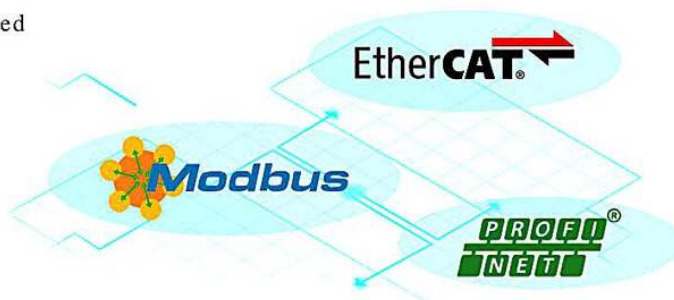
Powerful protection features

- Provides comprehensive protection for motors and Frequency Inverters, such as overcurrent, overvoltage, undervoltage, overheating, short circuit, motor stall, and motor overload.
- For pump applications, it provides a broken pipe protection, which can prioritize water supply in emergency situations.



Flexible I/O interfaces and communication capabilities

- It provides standard analog and digital input/output terminals, which can be connected to various sensors and control signals.
- With built-in Modbus RTU fieldbus communication protocol, it can be easily integrated into existing automation systems.
- It offers a variety of optional fieldbus adapter cards (such as Profibus-DP, EtherCAT, etc.) to expand communication flexibility.



Product technical specifications and specifications

Item		Item description
Input	Rated volt.,frequency	3 phase 380V, 50Hz/60z
	Allowed volt.range	3 phase 380V: 320-460V
Output	Voltage	0-input voltage
	Frequency	0~599Hz
	Over loading capacity	P type: 18.5kW and below: 120% of rated current for 1 minute 22kW-75kW: 115% of rated current for 1 minute 90kW and above: 110% of rated current for 1 minute
		G type: 150% of rated current for 1 minute
Control performance	Control mode	1. Without PG vector control(support synchronous motor and asynchronous motor) 2. PG vector control(support synchronous motor and asynchronous motor) 3. V/F control
	Speed control precision	±0.5% rated synchronous speed(without PG vector control) ±0.3% rated synchronous speed(PG vector control) ±1% rated synchronous speed(V/F control)
	Speed regulation range	1:2000(PG vector control) 1:200(without PG vector control) 1:50(V/F control)
	Starting torque	1.0Hz: 150% rated torque(V/F control) 0.25Hz: 150% rated torque(without PG vector control) 0Hz: 180% rated torque(PG vector control)
	Speed fluctuation	±0.3% rated synchronous speed(without PG vector control) ±0.1% rated synchronous speed(PG vector control)
	Torque response	≤20ms(without PG vector control) ≤10ms(PG vector control)
	Frequency precision	Digital setting: max.frequency×±0.01%,; Analog setting: max.frequency×±0.5%,
	Freq. resolution	Analog setting
		Digital setting
		External pulse
	Torque boost	Automatic torque boost Manual torque boost 0.1-30.0%
	V/F curve (volt. Frequency characteristics)	Setting rated frequency at the range of 5-599Hz, by choosing constant torque, degressive torque 1, degressive torque 2, degressive torque 3, self-defined V/F, V/F separation, total 6 kinds of curve
	Acceleration/Deceleration curve	Two modes: straight line acceleration and deceleration, S curve acceleration and deceleration; 15 kinds of acceleration and deceleration time, time unit(0.01s, 0.1s, 1s) for option, max. time for 1000 minutes
	Brake	Power consumption brake
		DC brake
	Jog	Jog frequency range: 0Hz - upper limit frequency; jog acceleration and deceleration time 0.1 - 6000.0 seconds for setting
	Multi-section speed run	Realized by inbuilt PLC or control terminal; with 15 section speed, each section speed with separately acceleration and deceleration time; with inbuilt PLC can achieve reserve when power down
	Inbuilt PID controller	Convenient to make closed-loop control system
	Automatic voltage(AVR)	Automatically keep output voltage constant, when the power grid voltage fluctuation
	Automatic current limiting	Current limited automatically under run mode in avoid of inverter over-current frequently to trip
	Carrier modulation	Modulate carrier wave automatically according to the load characteristic
	Speed tracking restart	Make rotating motor smoothly start without shocking
Running function	Running command specified channel	Keypad specified, control terminal specified, communication specified, can switch through various means
	Running frequency specified channel	Main & auxiliary specified to a realize one main adjusting and one fine control Digital specified, analog specified, pulse specified, pulse width specified, communication specified and others, which can be switched by many means at any time
	Binding function	Run command channel and frequency specified channel can bind together randomly and switch synchronously

Product technical specifications and specifications

Item		Item description
Input/output characteristic	Digital input channel	Channel 5 for universal digital input, max. frequency 1KHz, channel 1 can be used as pulse input channel, max. input 50KHz, expansion terminals can be realized
	Analog input channel	Channel 2 for analog input channel, AI1 is 0-10V input, AI2 can choose 4-20mA or 0-10V or 0-20mA input, expansion terminal can be realized
	Pulse output channel	0.1-20KHz pulse square signal output to achieve setting frequency, output frequency and other physical quantity output
	Analog output channel	Channel 1 for analog output channel, AO can choose 0-20mA or 0-10V to achieve setting frequency, output frequency and other physical quantity output, expansion terminal can be realized
	Monopulse control	Suitable for working site where need one button to control inverter start and stop, first press to start, then press to stop, and that cycle repeats. Its very simple and reliable
	Fixed length control	Realize fixed length control
	Timing control	Timing control function: setting time range 0.1Min - 6500.0Min
	Virtual terminal	Five group virtual input & output IO can realize simply logical control
Operating keypad	Keypad display	The parameters as setting frequency, output frequency, output voltage, output current can be displayed
	Button locked	Lock all or part of the buttons
Protection function		Motor power on short circuit test, input & output phase loss protection, over-current protection, over voltage protection, under voltage protection, over heat protection, overload protection, under load protection, short circuit protection to ground, terminal protection and no stop protection under power off, etc.
Environment	Application site	Indoor, not bare to sunlight, no dust, no corrosive gas, no flammable gas, no vapor, no water drop or salt etc.
	Altitude	Under 1000 meter(above 1000 meter require to reduce volume to use, output current reduce about 10% of rated current per 1000 meter high
	Environment temperature	-10°C - +40°C(environment temperature between 40°C - 50°C, need to reduce volume or strengthen heat sink)
	Environment humidity	Smaller than 95%RH, no drop condenses
	Vibration	Smaller than 5.9M/S ² (0.6g)
	Storage temperature	-40°C~+70°C
Structure	Protection grade	185kW and below: IP20; 200kW and above: IP00
	Cooling mode	Forced air cooling
Installation mode		185kW and below: wall-mounted; 200kW and above: cabinet-type, supporting cabinet installation

Inverter type description

EN760 - 4T 0015P / 0007G B

①	
Code	Name
EN760	Product Model

②	
Code	Volt. grade
4T	3 phase 380V

③	
Code	Fan and water pump special type motor power adapter(KW)
0015P	1.5
0022P	2.2
...	...
4500P	450
5000P	500

④	
Code	Heavy-duty general-purpose adaptable motor power(KW)
0007G	0.75
0015G	1.5
...	...
4000G	400
4500G	450

⑤	
Code	Name
B	Inbuilt brake unit

Inverter Series Model Description

Input voltage	Inverter model	Input current(A)	Rated output current (A)	Compatible motor
3 phase 380V	EN760-4T0015P/0007GB	5/4	3.7/2.9	1.5/0.75
	EN760-4T0022P/0015GB	7/5	5/4	2.2/1.5
	EN760-4T0037P/0022GB	11/9	8.5/6.6	3.7/2.2
	EN760-4T0055P/0037GB	17/14	13/10.5	5.5/3.7
	EN760-4T0075P/0055GB	22/18	17/13.5	7.5/5.5
	EN760-4T0110P/0075GB	33/26	25/20	11/7.5
	EN760-4T0150P/0110GB	43/33	33/25	15/11
	EN760-4T0185P/0150GB	51/43	39/33	18.5/15
	EN760-4T0220P/0185GB	59/51	45/39	22/18.5
	EN760-4T0300P/0220GB	78/62	60/47.5	30/22
	EN760-4T0370P/0300G	98/78	75/60	37/30
	EN760-4T0450P/0370G	118/98	91/75	45/37
	EN760-4T0550P/0450G	146/118	112/91	55/45
	EN760-4T0750P/0550G	195/153	150/118	75/55
	EN760-4T0900P/0750G	229/195	176/150	90/75
	EN760-4T1100P/0900G	273/229	210/176	110/90
	EN760-4T1320P/1100G	239/273	253/210	132/110
	EN760-4T1600P/1320G	288/239	304/253	160/132
	EN760-4T1850P/1600G	322/288	340/304	185/160
	EN760-4T2000P/1850G	365/288	380/340	200/160
	EN760-4T2200P/2000G	409/365	426/380	220/200
	EN760-4T2500P/2200G	441/409	474/426	250/220
	EN760-4T2800P/2500G	495/441	520/474	280/250
	EN760-4T3150P/2800G	570/495	600/520	315/280
	EN760-4T3550P/3150G	617/570	650/600	355/315
	EN760-4T3750P/3550G	646/617	680/650	375/355
	EN760-4T4000P/3750G	712/646	750/680	400/375
	EN760-4T4500P/4000G	760/712	800/750	450/400
	EN760-4T5000P/4500G	826/760	870/800	500/450

External shape and installation dimensions

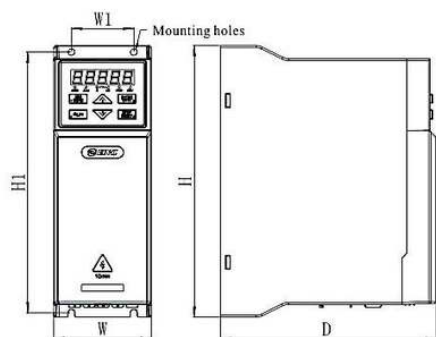


Fig.a

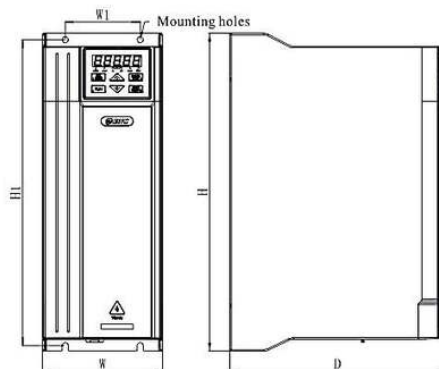


Fig.b

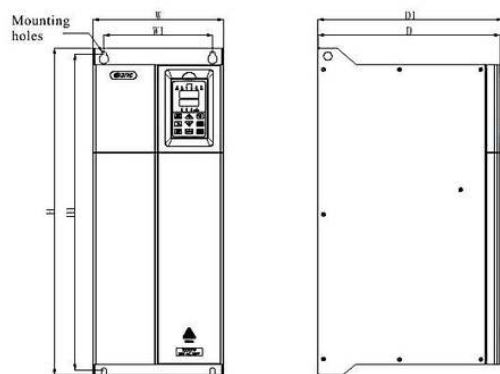


Fig.c

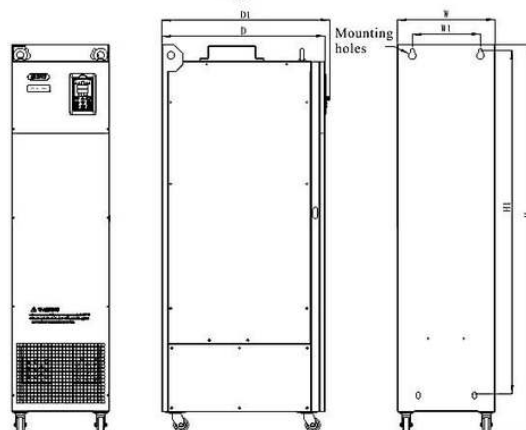


Fig.d

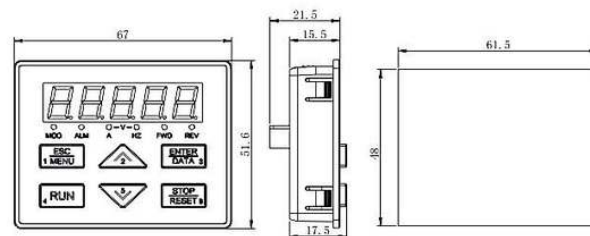
Input Voltage	W (mm)	H (mm)	D (mm)	W1 (mm)	H1 (mm)	D1 (mm)	Fix Hole (mm)	Fig.
EN760-4T0015P/0007GB	67	177	145	40	167	-	5	Fig.a
EN760-4T0022P/0015GB								
EN760-4T0037P/0022GB								
EN760-4T0055P/0037GB	75	210	166	48	201.5	-	5	Fig.a
EN760-4T0075P/0055GB	100	242	180	71	232.5	-	6	Fig.b
EN760-4T0110P/0075GB	120	320	214.7	75	308	-	6	Fig.b
EN760-4T0150P/0110GB								
EN760-4T0185P/0150GB	142	408	225	110	394	-	6	Fig.b
EN760-4T0220P/0185GB								
EN760-4T0300P/0220GB								
EN760-4T0370P/0300G	172	466	230	142	452	-	7	Fig.b
EN760-4T0450P/0370G								
EN760-4T0550P/0450G	225	550	315	185	530	319.5	7	Fig.c
EN760-4T0750P/0550G								
EN760-4T0900P/0750G	240	600	335	200	580	339.5	9	Fig.c
EN760-4T1100P/0900G								
EN760-4T1320P/1100G	310	730	390	230	710	394.5	9	Fig.c
EN760-4T1600P/1320G	335	750	405	245	730	409.5	9	Fig.c
EN760-4T1850P/1600G								
EN760-4T2000P/1850G	300	1150	505	180	1005	509.5	13	Fig.d

Input Voltage	W (mm)	H (mm)	D (mm)	W1 (mm)	H1 (mm)	D1 (mm)	Fix Hole (mm)	Fig.
EN760-4T2200P/2000G	300	1150	505	180	1005	509.5	13	Fig.d
EN760-4T2500P/2200G								
EN760-4T2800P/2500G								
EN760-4T3150P/2800G	330	1310	550	230	1165	554.5	13	Fig.d
EN760-4T3550P/3150G								
EN760-4T3750P/3550G								
EN760-4T4000P/3750G	340	1415	555	240	1270	559.5	13	Fig.d
EN760-4T4500P/4000G								
EN760-4T5000P/4500G								

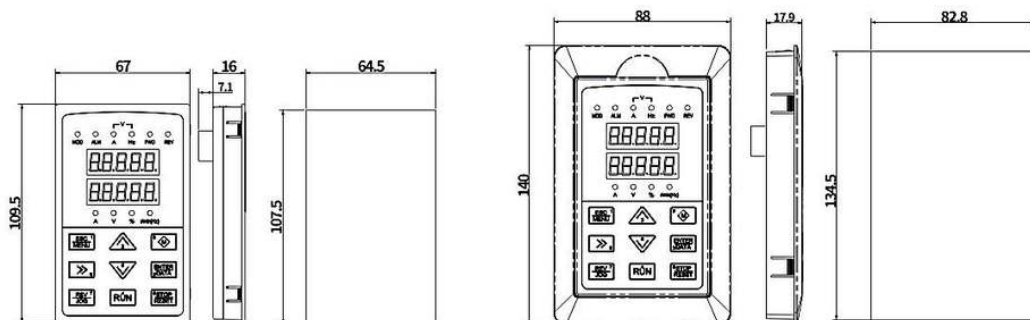
Three-phase input adapter copper busbar assembly (correspondence table between Frequency Inverter and three-phase input adapter copper busbar assembly)

Input Voltage	Three-phase input adapter copper busbar assembly model
EN760-4T2000P/1850G	SP-CTP-2200
EN760-4T2200P/2000G	
EN760-4T2500P/2200G	
EN760-4T2800P/2500G	SP-CTP-2800
EN760-4T3150P/2800G	
EN760-4T3550P/3150G	
EN760-4T3750P/3550G	SP-CTP-4000
EN760-4T4000P/3750G	
EN760-4T4500P/4000G	
EN760-4T5000P/4500G	

Operating Keypad and Keypad Mounting Box Outline Dimensions
(unit: mm)



◆ EN-LED15 Keyboard Shape and Cutout Dimensions



◆ EN-LED16 Keyboard Shape and Cutout Dimensions

◆ EN-LED16 Keyboard Mounting Box Shape and Cutout Dimensions

- Note:**
- (1) When using the EN-LED15 or EN-LED15-D keypad for remote mounting, it cannot be installed using the keypad mounting box; only the keypad itself can be installed. Please refer to the diagram for the cutout dimensions.
 - (2) When using the EN-LED16 or EN-LED16-D keypad for remote mounting, customers can choose the cutout dimensions based on either the keypad or the keypad mounting box according to actual needs. It is recommended that the mounting panel thickness be between 1.0 mm and 1.5 mm.
 - (3) If installation requires the keypad mounting box, it must be ordered separately.